



Analytical Laboratory Report

Report ID: S73932.01(01)
Generated on 05/15/2025

Report to

Attention: Cindy Tomaszewski
City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368 FAX:
Email: lab@cadillac-mi.net

Report produced by

Merit Laboratories, Inc.
2680 East Lansing Drive
East Lansing, MI 48823

Phone: (517) 332-0167 FAX: (517) 332-6333

Contacts for report questions:
John Lavery (johnlavery@meritlabs.com)
Barbara Ball (bball@meritlabs.com)

Report Summary

Lab Sample ID(s): S73932.01
Project: Quarterly Biosolids
Collected Date(s): 04/28/2025
Submitted Date/Time: 04/29/2025 10:00
Sampled by: Unknown
P.O. #:

Table of Contents

- Cover Page (Page 1)
- General Report Notes (Page 2)
- Report Narrative (Page 2)
- Laboratory Accreditations (Page 3)
- Qualifier Descriptions (Page 3)
- Glossary of Abbreviations (Page 3)
- Method Summary (Page 4)
- Parameter Summary (Page 5)
- Sample Summary (Page 6)

Maya Murshak
Technical Director



Analytical Laboratory Report

General Report Notes

Analytical results relate only to the samples tested, in the condition received by the laboratory.

Methods may be modified for improved performance.

Results reported on a dry weight basis where applicable.

'Not detected' indicates that parameter was not found at a level equal to or greater than the reporting limit (RL).

When MDL results are provided, then 'Not detected' indicates that parameter was not found at a level equal to or greater than the MDL.

40 CFR Part 136 Table II Required Containers, Preservation Techniques and Holding Times for the Clean Water Act specify that samples for acrolein and acrylonitrile, and 2-chloroethylvinyl ether need to be preserved at a pH in the range of 4 to 5 or if not preserved, analyzed within 3 days of sampling.

QA/QC corresponding to this analytical report is a separate document with the same Merit ID reference and is available upon request.

Starred (*) analytes are not NY NELAP accredited.

Samples are held by the lab for 30 days from the final report date unless a written request to hold longer is provided by the client.

Report shall not be reproduced except in full, without the written approval of Merit Laboratories, Inc.

Limits for drinking water samples, are listed as the MCL Limits (Maximum Contaminant Level Concentrations)

PFAS requirement: Section 9.3.8 of U.S. EPA Method 537.1 states "If the method analyte(s) found in the Field Sample is present in the FRB at a concentration greater than 1/3 the MRL, then all samples collected with that FRB are invalid and must be recollected and reanalyzed."

Samples submitted without an accompanying FRB may not be acceptable for compliance purposes.

Wisconsin PFAs analysis: MDL = LOD; RL = LOQ. LOD and LOQ are adjusted for dilution.

All accreditations/certifications held by this laboratory are listed on page 3. Not all accreditations/certifications are applicable to this report.

For a specific list of accredited analytes, please feel free to contact the laboratory or visit <https://www.meritlabs.com/certifications>.

Report Narrative

There is no additional narrative for this analytical report

Laboratory Accreditations (For Reference Only)

Authority	Accreditation ID
Michigan DEQ	#9956
DOD ELAP & ISO/IEC 17025:2017	#69699 PJLA Testing
WBENC	#2005110032
Ohio VAP	#CL0002
Indiana DOH	#C-MI-07
New York NELAC	#11814
North Carolina DENR	#680
North Carolina DOH	#26702
Pennsylvania DEP	#68-05884
Wisconsin DNR	FID# 399147320

Qualifier Descriptions

Qualifier	Description
!	Result is outside of stated limit criteria
B	Compound also found in associated method blank
E	Concentration exceeds calibration range
F	Analysis run outside of holding time
G	Estimated result due to extraction run outside of holding time
H	Sample submitted and run outside of holding time
I	Matrix interference with internal standard
J	Estimated value less than reporting limit, but greater than MDL
L	Elevated reporting limit due to low sample amount
M	Result reported to MDL not RDL
O	Analysis performed by outside laboratory. See attached report.
R	Preliminary result
S	Surrogate recovery outside of control limits
T	No correction for total solids
X	Elevated reporting limit due to matrix interference
Y	Elevated reporting limit due to high target concentration
b	Value detected less than reporting limit, but greater than MDL
e	Reported value estimated due to interference
j	Analyte also found in associated method blank
o	Associated EIS outside of control limits
p	Benzo(b)Fluoranthene and Benzo(k)Fluoranthene integrated as one peak.
q	Qualifier ion ratio outside of control limits
x	Preserved from bulk sample

Glossary of Abbreviations

Abbreviation	Description
RL/RDL	Reporting Limit
MDL	Method Detection Limit
MS	Matrix Spike
MSD	Matrix Spike Duplicate
SW	EPA SW 846 (Soil and Wastewater) Methods
E	EPA Methods
SM	Standard Methods
LN	Linear
BR	Branched



Analytical Laboratory Report

Method Summary

Method	Version
EPA 1633A	EPA Method 1633 Revision A December 2024
SM2540B	Standard Method 2540 B 2020

Parameter Summary

Parameter	Synonym	Cas #
PFBA	Perfluorobutanoic Acid	375-22-4
PFPeA	Perfluoropentanoic Acid	2706-90-3
4:2 FTSA	4:2 Fluorotelomer Sulfonic Acid	757124-72-4
PFHxA	Perfluorohexanoic Acid	307-24-4
PFBS	Perfluorobutane sulfonic Acid	375-73-5
PFHpA	Perfluoroheptanoic Acid	375-85-9
PFPeS	Perfluoropentane Sulfonic Acid	2706-91-4
6:2 FTSA	6:2 Fluorotelomer Sulfonic Acid	27619-97-2
PFOA	Perfluorooctanoic Acid	335-67-1
PFHxS	Perfluorohexane Sulfonic Acid	355-46-4
PFNA	Perfluorononanoic Acid	375-95-1
8:2 FTSA	8:2 Fluorotelomer Sulfonic Acid	39108-34-4
PFHpS	Perfluoroheptane Sulfonic Acid	375-92-8
PFDA	Perfluorodecanoic Acid	335-76-2
N-MeFOSAA	N-methyl perfluorooctanesulfonamidoacetic acid	2355-31-9
EtFOSAA	N-Ethyl Perfluorooctane Sulfonamidoacetic Acid	2991-50-6
PFOS	Perfluorooctane Sulfonic Acid	1763-23-1
PFUnDA	Perfluoroundecanoic Acid	2058-94-8
PFNS	Perfluorononane Sulfonic Acid	68259-12-1
PFDoDA	Perfluorododecanoic Acid	307-55-1
PFDS	Perfluorodecane Sulfonic Acid	335-77-3
PFTTrDA	Perfluorotridecanoic Acid	72629-94-8
FOSA	Perfluorooctane Sulfonamide	754-91-6
PFTeDA	Perfluorotetradecanoic Acid	376-06-7
11Cl-PF3OUdS	11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051-92-9
9Cl-PF3ONS	9-chlorohexadecafluoro-3-oxanone1-sulfonic acid	756426-58-1
ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
HFPO-DA	Hexafluoropropylene oxide dimer	13252-13-6
FHpPA (7:3 FTCA)	3-Perfluoroheptyl propanoic acid	812-70-4
FPePA (5:3 FTCA)	3-Perfluoropentyl propanoic acid	914637-49-3
FPrPA (3:3 FTCA)	3-Perfluoropropyl propanoic acid	356-02-5
NFDHA	Nonafluoro-3,6-dioxaheptanoic acid	151772-58-6
PFEESA	Perfluoro(2-ethoxyethane)sulfonic acid	113507-82-7
PFMBA	Perfluoro-4-methoxybutanoic acid	863090-89-5
PFMPA	Perfluoro-3-methoxypropanoic acid	377-73-1
NMeFOSAM	N-Methylperfluorooctanesulfonamide	31506-32-8
NMeFOSE	N-Methylperfluorooctanesulfonamidoethanol	24448-09-7
NEtFOSAM	N-Ethylperfluorooctanesulfonamide	4151-50-2
NEtFOSE	N-Ethylperfluorooctanesulfonamidoethanol	1691-99-2
PFDoS	Perfluorododecanesulfonic acid	79780-39-5



Analytical Laboratory Report

Sample Summary (1 samples)

Sample ID	Sample Tag	Matrix	Collected Date/Time
S73932.01	Digester #4	Sludge	04/28/25 12:54



Analytical Laboratory Report

Lab Sample ID: S73932.01

Sample Tag: Digester #4

Collected Date/Time: 04/28/2025 12:54

Matrix: Sludge

COC Reference: 175249

Sample Containers

#	Type	Preservative(s)	Refrigerated?	Arrival Temp. (C)	Thermometer #
1	250mL Plastic	None	Yes	4.4	IR

Extraction / Prep.

Parameter	Result	Method	Run Date	Analyst	Flags
Initial wt. (g) / Final wt. (g) / Volume (ml)*	32.99/30/5	EPA 1633A	05/07/25 10:00	KYD	
Initial wt. (g) / Final wt. (g) / Volume (ml) (Reps)	50.18/30/5	EPA 1633A	05/12/25 12:00	KYD	

Inorganics

Method: SM2540B, Run Date: 04/30/25 14:15, Analyst: MAM

Parameter	Result	RL	MDL	Units	Dilution	CAS#	Flags
Total Solids*	1.3	1		%	1		

Organics

PFA's EPA Method 1633, Method: EPA 1633A, Run #1: 05/13/25 14:17, #2: 05/08/25 16:10, Analyst: KYD

Parameter	Result	RL	MDL	Units	Dilution	Run #	Flags
PFBA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFPeA*	Not detected	1.9		ug/kg	19.1	Run #1	r
4:2 FTSA*	Not detected	1.9		ug/kg	19.1	Run #1	or
PFHxA*	4.9	1.9		ug/kg	19.1	Run #1	r
PFBS*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFHpA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFPeS*	Not detected	1.9		ug/kg	19.1	Run #1	r
6:2 FTSA*	Not detected	1.9		ug/kg	19.1	Run #1	or
PFOA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFHxS*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFNA*	Not detected	1.9		ug/kg	19.1	Run #1	r
8:2 FTSA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFHpS*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFDA*	Not detected	1.9		ug/kg	19.1	Run #1	r
N-MeFOSAA*	11	1.9		ug/kg	19.1	Run #1	r
EtFOSAA*	3	1.9		ug/kg	19.1	Run #1	r
PFOS*	3.9	1.9		ug/kg	19.1	Run #1	r
PFUnDA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFNS*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFDoDA*	Not detected	13		ug/kg	129	Run #2	*r
PFDS*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFTTrDA*	Not detected	13		ug/kg	129	Run #2	*r
FOSA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFTeDA*	Not detected	13		ug/kg	129	Run #2	*r
11Cl-PF3OUdS*	Not detected	1.9		ug/kg	19.1	Run #1	r
9Cl-PF3ONS*	Not detected	1.9		ug/kg	19.1	Run #1	r
ADONA*	Not detected	1.9		ug/kg	19.1	Run #1	r
HFPO-DA*	Not detected	1.9		ug/kg	19.1	Run #1	r
FHpPA (7:3 FTCA)*	24	3.8		ug/kg	19.1	Run #1	r

r-This analyte is being reported as the best result from multiple runs

o-Associated EIS outside of control limits



Analytical Laboratory Report

Lab Sample ID: S73932.01 (continued)

Sample Tag: Digester #4

PFA's EPA Method 1633, Method: EPA 1633A, Run #1: 05/13/25 14:17, #2: 05/08/25 16:10, Analyst: KYD (continued)

Parameter	Result	RL	MDL	Units	Dilution	Run #	Flags
FPePA (5:3 FTCA)*	330	3.8		ug/kg	19.1	Run #1	r
FPrPA (3:3 FTCA)*	Not detected	3.8		ug/kg	19.1	Run #1	r
NFDHA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFEESA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFMBA*	Not detected	1.9		ug/kg	19.1	Run #1	r
PFMPA*	Not detected	1.9		ug/kg	19.1	Run #1	r
NMeFOSAM*	Not detected	1.9		ug/kg	19.1	Run #1	r
NMeFOSE*	9.9	1.9		ug/kg	19.1	Run #1	r
NEtFOSAM*	Not detected	1.9		ug/kg	19.1	Run #1	r
NEtFOSE*	2.9	1.9		ug/kg	19.1	Run #1	r
PFDoS*	Not detected	13		ug/kg	129	Run #2	*r

r-This analyte is being reported as the best result from multiple runs

Merit Laboratories Login Checklist

Lab Set ID:S73932

Client:CADILLAC (City of Cadillac - WWTP)

Project: Quarterly Biosolids

Submitted:04/29/2025 10:00 Login User: JEH

Attention: Cindy Tomaszewski

Address: City of Cadillac - WWTP
1121 Plett Road
Cadillac, MI 49601

Phone: 231-775-2368

FAX:

Email: lab@cadillac-mi.net

Selection	Description	Note
Sample Receiving		
01.	☒ Yes ☐ No ☐ N/A	Samples are received at 4C +/- 2C Thermometer # IR: 4.4C
02.	☒ Yes ☐ No ☐ N/A	Received on ice/ cooling process begun
03.	☒ Yes ☐ No ☐ N/A	Samples shipped UPS
04.	☐ Yes ☒ No ☐ N/A	Samples left in 24 hr. drop box
05.	☒ Yes ☐ No ☐ N/A	Are there custody seals/tape or is the drop box locked
Chain of Custody		
06.	☒ Yes ☐ No ☐ N/A	COC adequately filled out
07.	☒ Yes ☐ No ☐ N/A	COC signed and relinquished to the lab
08.	☒ Yes ☐ No ☐ N/A	Sample tag on bottles match COC
09.	☐ Yes ☒ No ☐ N/A	Subcontracting needed? Subcontacted to:
Preservation		
10.	☒ Yes ☐ No ☐ N/A	Do sample have correct chemical preservation
11.	☐ Yes ☐ No ☒ N/A	Completed pH checks on preserved samples? (no VOAs)
12.	☐ Yes ☒ No ☐ N/A	Did any samples need to be preserved in the lab?
Bottle Conditions		
13.	☒ Yes ☐ No ☐ N/A	All bottles intact
14.	☒ Yes ☐ No ☐ N/A	Appropriate analytical bottles are used
15.	☒ Yes ☐ No ☐ N/A	Merit bottles used
16.	☒ Yes ☐ No ☐ N/A	Sufficient sample volume received
17.	☐ Yes ☒ No ☐ N/A	Samples require laboratory filtration
18.	☒ Yes ☐ No ☐ N/A	Samples submitted within holding time
19.	☐ Yes ☐ No ☒ N/A	Do water VOC, TOX, DO or Alkalinity bottles contain

Corrective action for all exceptions is to call the client and to notify the project manager.

Client Review By: _____ Date: _____

REPORT TO

CHAIN OF CUSTODY RECORD

INVOICE TO

CONTACT NAME CINDY TOMASZEWSKI			
COMPANY City of Cadillac - WWTP			
ADDRESS 1121 Plett Rd			
CITY Cadillac		STATE MI	ZIP CODE 49601
PHONE NO. 231-775-2368		CELL NO.	
E-MAIL ADDRESS lab@cadillac-mi.net		P.O. NO.	
		QUOTE NO.	

CONTACT NAME		CINDY TOMASZEWSKI		SAME	
COMPANY					
ADDRESS					
CITY				STATE	ZIP CODE
PHONE NO.		E-MAIL ADDRESS			

PROJECT NO./NAME QUARTERLY BIOSOLIDS	SAMPLER(S) - PLEASE PRINT/SIGN NAME
TURNAROUND TIME REQUIRED ☐ 1 DAY ☐ 2 DAYS ☐ 3 DAYS ☒ STANDARD ☐ OTHER _____	
DELIVERABLES REQUIRED ☒ LEVEL II ☐ LEVEL III ☐ LEVEL IV ☐ EDD ☐ OTHER _____	

MATRIX	W=WATER	GW=GROUNDWATER	WW=WASTEWATER	S=SOIL	L=LIQUID	SD=SOLID
CODE:	SL=SLUDGE	DW=DRINKING WATER	O=OIL	WP=WIPE	A=AIR	WS=WASTE

[illegible]

ANALYSIS (ATTACH LIST IF MORE SPACE IS REQUIRED)									
# Containers & Preservatives									
NONE	HCl	HNO ₃	H ₂ SO ₄	NaOH	MeOH	OTHER			
X							X	PFAS - MHD. 1633	
							X	Total Solids	
							Certifications ☐ OHIO VAP ☐ Drinking Water ☐ DoD ☒ NPDES Project Locations ☐ Detroit ☐ New York ☒ Other <u>MICHIGAN</u> Special Instructions <u>EGLE Param. Lis</u> <u>3 RL's</u>		

RELINQUISHED BY: SIGNATURE/ORGANIZATION	<i>T. Loggins / City of Cadillac</i>	DATE	TIME
RECEIVED BY: SIGNATURE/ORGANIZATION		4-28-25	12
RELINQUISHED BY: SIGNATURE/ORGANIZATION		DATE	TIME
RECEIVED BY: SIGNATURE/ORGANIZATION		DATE	TIME

RELINQUISHED BY:	UPS	DATE	TIME
SIGNATURE/ORGANIZATION		4-27-05	1000
RECEIVED BY:	J. Helms	DATE	TIME
SIGNATURE/ORGANIZATION		4-27-05	1000

SEAL NO.	SEAL INTACT	INITIALS	NOTES:
	YES ☐ NO ☐		TEMP. ON ARRIVAL _____
SEAL NO.	SEAL INTACT	INITIALS	
	YES ☐ NO ☐		4.4°C